

ALERT AP, NU, Canada

WMO: 710820

Lat: 82.520N

Lon: 62.280W

Elev: 31

StdP: 100.96

Time Zone: -5.00 (NAE)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.2	-41.7	-44.7	0.0	-41.0	-44.1	0.0	-40.5	18.8	-21.4	16.3	-22.2	0.9	300	1.166

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	11.9	7.4	9.1	5.8	6.7	4.2	7.5	11.5	5.9	9.0	4.3	6.5	9.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.1	5.1	8.7	2.9	4.7	7.1	2.0	4.4	4.9	24.0	11.7	20.6	9.1	17.4	6.6	12.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.5	10.2	DB	-45.2	15.4	2.1	1.8	-46.7	16.7	-48.0	17.7	-49.1	18.7	-50.7	20.0	(4)
(5)			WB	-44.4	9.4	1.1	1.7	-45.2	10.7	-45.8	11.7	-46.4	12.7	-47.2	13.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-17.5	-31.7	-33.2	-32.7	-24.2	-11.7	-0.8	3.1	0.6	-8.4	-18.2	-24.9	-29.4	(6)	
	DBStd	14.04	5.33	5.32	5.60	6.23	5.00	3.25	2.86	3.42	5.07	5.30	5.52	5.47	(7)	
	HDD10.0	10046	1293	1211	1323	1026	673	323	213	292	552	874	1046	1221	(8)	
	HDD18.3	13087	1551	1444	1581	1276	931	573	471	550	802	1132	1296	1479	(9)	
	CDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.6	2.6	2.4	1.8	2.2	2.5	2.7	3.4	3.2	3.1	2.8	2.5	2.5	(14)	
(15) Precipitation	PrecAvg	172	8	6	8	10	12	13	29	25	27	15	11	8	(15)	
	PrecMax	292	33	22	27	38	28	47	57	59	59	42	26	23	(16)	
	PrecMin	94	2	0	0	1	1	0	3	2	2	4	0	2	(17)	
	PrecStd	36	5	5	5	9	8	11	15	14	14	10	7	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-15.5	-12.2	-12.9	-4.9	2.1	10.9	14.4	13.3	5.0	-2.3	-7.0	-11.8	(19)	
		MCWB	-16.1	-12.4	-13.3	-6.2	0.5	6.9	9.2	8.2	2.1	-4.1	-8.2	-12.7	(20)	
	2%	DB	-19.0	-20.1	-17.0	-8.7	-0.9	7.2	12.8	9.4	2.2	-5.8	-9.9	-14.2	(21)	
		MCWB	-19.7	-20.7	-17.3	-9.4	-2.3	4.2	8.0	5.7	0.1	-7.0	-10.6	-14.8	(22)	
	5%	DB	-21.4	-22.9	-21.3	-12.0	-3.0	5.1	11.3	7.4	-0.4	-9.0	-13.3	-17.4	(23)	
		MCWB	-21.8	-23.2	-21.7	-12.5	-3.8	3.2	7.0	4.4	-1.5	-9.7	-14.0	-18.0	(24)	
	10%	DB	-23.3	-25.5	-24.4	-15.0	-5.0	3.4	8.4	5.4	-1.8	-10.7	-16.5	-20.6	(25)	
		MCWB	-23.7	-25.9	-24.7	-15.5	-5.7	2.1	5.6	3.0	-2.6	-11.3	-17.0	-21.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-15.9	-12.4	-13.1	-5.9	0.7	7.0	9.7	8.6	2.2	-3.8	-7.9	-12.7	(27)	
		MCDB	-15.5	-12.3	-12.6	-4.7	2.0	10.1	13.9	12.0	4.6	-2.4	-7.1	-12.0	(28)	
	2%	WB	-19.2	-20.6	-17.0	-9.2	-2.1	4.5	8.4	5.9	0.2	-6.9	-10.6	-14.7	(29)	
		MCDB	-18.7	-20.3	-16.5	-8.6	-1.1	6.9	12.7	9.3	1.8	-5.8	-9.8	-14.1	(30)	
	5%	WB	-21.9	-23.1	-21.3	-12.4	-3.9	3.2	7.1	4.5	-1.4	-9.7	-13.9	-17.7	(31)	
		MCDB	-21.5	-22.8	-21.0	-11.9	-3.2	5.0	11.0	7.2	-0.5	-9.2	-13.3	-17.2	(32)	
	10%	WB	-24.0	-25.8	-24.2	-15.3	-5.6	2.1	5.7	3.1	-2.6	-11.2	-16.9	-20.9	(33)	
		MCDB	-23.7	-25.5	-23.8	-14.8	-5.0	3.4	8.6	5.0	-1.8	-10.7	-16.5	-20.5	(34)	
(35) Mean Daily Temperature Range	MDBR		6.8	6.9	6.9	6.7	4.8	3.6	4.7	4.3	5.2	6.5	6.6	6.6	(35)	
	5% DB	MCDBR	9.3	10.9	11.8	9.5	6.8	6.5	10.3	8.3	5.7	8.8	9.7	10.4	(36)	
		MCWBR	9.8	10.8	11.5	9.0	5.9	4.4	6.5	5.6	4.6	8.0	9.3	10.3	(37)	
	5% WB	MCDBR	10.2	10.9	12.2	9.4	6.3	6.3	9.9	8.3	5.5	8.6	9.8	10.6	(38)	
		MCWBR	9.9	10.7	11.9	9.0	5.5	4.4	6.5	5.6	4.5	7.9	9.4	10.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	N/A	0.174	0.252	0.274	0.300	0.291	0.269	0.194	N/A	N/A	N/A	(40)	
	taud		N/A	N/A	1.972	2.118	2.171	2.227	2.390	2.403	2.065	N/A	N/A	N/A	(41)	
	Ebn at Noon		0	0	504	736	813	808	805	740	486	0	0	0	(42)	
	Edh at Noon		0	0	31	73	91	93	75	57	30	0	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.00	0.00	0.52	3.41	6.28	7.17	6.03	3.34	0.95	0.00	0.00	0.00	(44)	
	RadStd		0.00	0.00	0.05	0.19	0.35	0.34	0.43	0.34	0.09	0.00	0.00	0.00	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page